

ENERGY

CHLOROPLAST

BASIC

E

P

COST

ACTIVATE

CONVERTER

$1 \text{ WATER} + 1 \text{ CO}_2 \rightarrow 2 \text{ GLUCOSE}$

Prereq: Light + water + CO₂
 BIOLOGY Makes glucose by photosynthesis.

CHL-01 ×4

ENERGY

CHLOROPLAST

LIGHT-HARVESTING

P

×2

COST

ACTIVATE

CONVERTER

Water + CO₂ → glucose; first Chloroplast in row: +1 glucose.

Prereq: Light + water + CO₂
 BIOLOGY Captures light energy to support photosynthesis.

CHL-LH

ENERGY

CHLOROPLAST

WATER-EFFICIENT

E

×2

COST

ACTIVATE

CONVERTER

If Vacuole stores ≥2 water: 1 CO₂ → 2 glucose.

Prereq: Vacuole stores ≥2 water
 BIOLOGY Links water storage to photosynthesis efficiency.

CHL-WE

ENERGY

CHLOROPLAST

NETWORK

P

G

COST

ACTIVATE

CONVERTER

+1 glucose per other adjacent Chloroplast; max +2.

Prereq: Adjacent Chloroplast
 BIOLOGY Multiple chloroplasts cooperate in a shared photosynthetic engine.

CHL-NET

ENERGY

MITOCHONDRION

BASIC

E

P

COST

ACTIVATE

CONVERTER

$1 \text{ GLUCOSE} \rightarrow 2 \text{ ENERGY}$

Prereq: Glucose available
 BIOLOGY Releases usable energy through cellular respiration.

MIT-01 ×4

ENERGY

MITOCHONDRION

EFFICIENT

E

P

COST

ACTIVATE

CONVERTER

Convert glucose → 2 energy, up to twice.

Prereq: Glucose available
 BIOLOGY Cellular respiration can run twice when glucose is available.

MIT-EF

ENERGY

MITOCHONDRION

CHLOROPLAST PARTNER

P

G

COST

ACTIVATE

CONVERTER

Glucose → 2 energy; Chloroplast glucose earlier: +1 energy.

Prereq: Chloroplast in this row
 BIOLOGY Links chloroplast glucose to cellular respiration.

MIT-CP

ENERGY

MITOCHONDRION

ENERGY DISTRIBUTOR

P

×2

COST

ACTIVATE

CONVERTER

Gain 2 energy; share 1 energy to gain +1 energy.

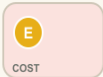
Prereq: None
 BIOLOGY Shares usable energy between cells.

MIT-DI

PROTEIN

RIBOSOME

FREE



ACTIVATE  CONVERTER

1 AMINO + 1 ENERGY → 1 PROTEIN

Prereq: Amino acids + energy

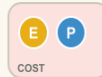
BIOLOGY Builds proteins from amino acids.

RIB-01 ×4

PROTEIN

RIBOSOME

CLUSTER



ACTIVATE  CONVERTER

Amino + energy → protein, up to twice.

Prereq: Amino acids + energy

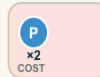
BIOLOGY A ribosome cluster can translate more than one set.

RIB-CL

PROTEIN

RIBOSOME

ER-BOUND



ACTIVATE  CONVERTER

With Rough ER directly right: amino + energy → 2 protein.

Prereq: Rough ER directly right

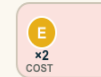
BIOLOGY ER-bound ribosomes make proteins for processing.

RIB-ER

PROTEIN

RIBOSOME

SHARED TRANSLATION



ACTIVATE  CONVERTER

Another player may pay amino to gain protein; you gain protein too.

Prereq: Another player has amino acids

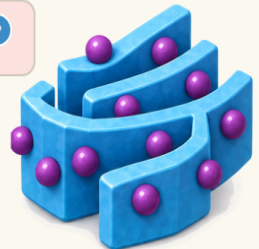
BIOLOGY Cells can share translation work.

RIB-SH

PROTEIN

ROUGH ER

PROTEIN PROCESSING



ACTIVATE  PROCESSOR

Protein made earlier in this row → +1 protein.

Prereq: Ribosome in this row

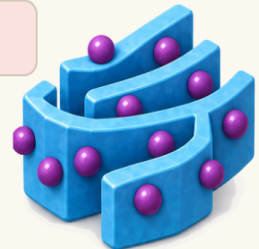
BIOLOGY Processes proteins made by ribosomes.

ER-01 ×3

PROTEIN

ROUGH ER

TRANSPORT



ACTIVATE  PROCESSOR

Move 1 protein onto this card; it counts as Processed Protein.

Prereq: Processed protein

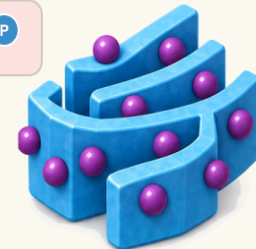
BIOLOGY Moves proteins through the endomembrane system.

ER-TR

PROTEIN

ROUGH ER

RIBOSOME DOCKING



PASSIVE  PROCESSOR

Ribosome immediately left produces +1 protein.

Prereq: Ribosome directly left

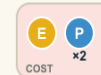
BIOLOGY Attached ribosomes increase local protein output.

ER-DO

PROTEIN

ROUGH ER

NETWORK



ACTIVATE  PROCESSOR

For every 2 Ribosomes in row, gain 1 protein; max 2.

Prereq: At least 2 Ribosomes

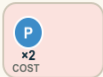
BIOLOGY A connected ER network supports several ribosomes.

ER-NT

PROTEIN

GOLGI APPARATUS

PACKAGING



ACTIVATE  PROCESSOR

1 processed protein → 1 Packaged Protein.

Prereq: Processed protein

BIOLOGY Sorts and packages proteins for transport.

GOL-01 x3

PROTEIN

GOLGI APPARATUS

SORTING



ACTIVATE  PROCESSOR

Package 1 protein; choose energy, draw 1, or basic nutrient.

Prereq: Processed protein

BIOLOGY Sorts proteins to different destinations.

GOL-00

PROTEIN

GOLGI APPARATUS

EFFICIENT



ACTIVATE  PROCESSOR

Package up to 2 protein.

Prereq: Two processed proteins

BIOLOGY Packages multiple proteins in one pass.

GOL-EF

PROTEIN

GOLGI APPARATUS

PROTEIN DELIVERY



ACTIVATE  PROCESSOR

Give 1 Packaged Protein; gain 2 energy + draw 1 card.

Prereq: Another player can receive protein

BIOLOGY Exports packaged proteins to another cell.

GOL-DE

SUPPORT

CENTRAL VACUOLE

WATER



BUILD + ACTIVATE  STORAGE

Store water here • water cap 5.

Prereq: Water available

BIOLOGY Stores water and maintains internal pressure.

VAC-01 x2

SUPPORT

CENTRAL VACUOLE

TURGOR



ACTIVATE  STORAGE

If this card stores ≥3 water: gain 1 VP / Growth marker.

Prereq: Vacuole stores ≥3 water

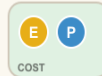
BIOLOGY Water pressure supports plant structure.

VAC-TU

SUPPORT

CENTRAL VACUOLE

STORAGE



PASSIVE  STORAGE

Store any 3 nutrients here; they bypass normal storage.

Prereq: None

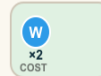
BIOLOGY Stores nutrients and other cell materials.

VAC-ST

SUPPORT

CENTRAL VACUOLE

WATER EXCHANGE



ACTIVATE  STORAGE

Give another player 1 water, then gain 2 water.

Prereq: Another player has water

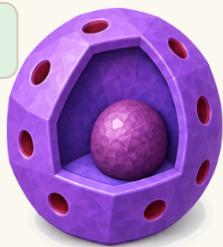
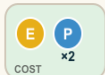
BIOLOGY Shares water while retaining more for the active cell.

VAC-EX

SUPPORT

NUCLEUS

GENETIC INSTRUCTIONS



BUILD + ACTIVATE



CONTROL

Draw 2 organelle cards; keep 1.

Prereq: None

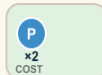
BIOLOGY Holds DNA instructions and guides cell activity.

NUC-G1 x2

SUPPORT

NUCLEUS

GENE SELECTION



ACTIVATE



CONTROL

Look at the top 3 cards; keep 1.

Prereq: None

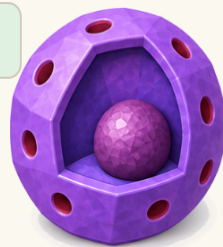
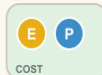
BIOLOGY Selects genetic instructions for cell activity.

NUC-GS

SUPPORT

NUCLEUS

CELL CONTROL



ACTIVATE



CONTROL

Choose another organelle in this row and activate it again.

Prereq: Another organelle in this row

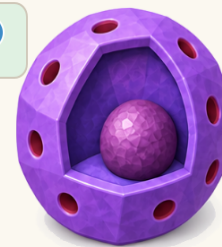
BIOLOGY Coordinates repeated organelle activity.

NUC-CC

SUPPORT

NUCLEUS

SHARED INSTRUCTIONS



ACTIVATE



CONTROL

Another player draws 1; you draw 2 and keep 1.

Prereq: Another player has a card

BIOLOGY Shares control information between cells.

NUC-SI

SUPPORT

CELL WALL

STRUCTURAL



PASSIVE



STRUCTURE

+1 point for each 3 organelles in your cell.

Prereq: Cell membrane present

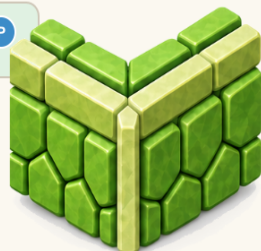
BIOLOGY Provides rigid support around a plant cell.

WAL-G1 x2

SUPPORT

CELL WALL

STRONG



PASSIVE



STRUCTURE

Once per phase, ignore one negative effect or resource loss.

Prereq: Cell membrane present

BIOLOGY Protects the cell from loss or stress.

WAL-ST

SUPPORT

CELL WALL

TURGOR SUPPORT



PASSIVE



STRUCTURE

Vacuole stores ≥ 3 water: +1 VP at phase end.Prereq: Vacuole stores ≥ 3 water

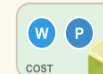
BIOLOGY Works with the vacuole to support turgor.

WAL-TU

SUPPORT

CELL WALL

EXPANDING



WHEN BUILT



STRUCTURE

Hand limit and row capacity +1.

Prereq: None

BIOLOGY Supports growth by increasing cell capacity.

WAL-EX

SUPPORT

CELL MEMBRANE

SELECTIVE



PASSIVE



UPGRADE

When taking a nutrient, choose 1 of 2 options.

Prereq: Cell membrane present

BIOLOGY Controls which nutrients enter the cell.

MEM-SEL • TBD

SUPPORT

CELL MEMBRANE

TRANSPORT PROTEIN



ACTIVATE



UPGRADE

Gain any 1 basic nutrient.

Prereq: Cell membrane present

BIOLOGY Moves a basic nutrient across the membrane.

MEM-TP • TBD